

Liu Yuchen

Phone: 19952517681

Email: yuchen.liu@connect.um.edu.mo

WeChat: LiuYC-091502

Location: Zhuhai / Macao

DOB: Sep. 2002

Gender: Male

Origin: Changzhou, Jiangsu

Research Interest: Public Health, Computational Social Science, Agent-Based Modeling, Multi-Omics Analysis

Education

Data Science (Precision Medicine), Faculty of Health Science, University of Macau Sep 2025 – Jun 2027

GPA: 3.44 **Research Direction:** Biostatistics, Epidemiology and Public Health

Supervisor: Prof. Luo Jiajun

Core Courses: Data Science Programming Methods (A), Data Analysis and Visualization (A-), Artificial Intelligence Methods in Biomedicine (A-), Medical and Healthcare Data Analysis, Practical Machine Learning (A-)

Activities: Media Center of UM Postgraduate Association; Volunteer for the 15th National Games of China

Sociology, School of Philosophy and Social Development, Huaqiao University Sep 2021 – Jun 2025

Average Score: 81.15/100

Selected Courses: Big Data and Computational Social Science (92), Urban Sociology (91), Social Survey and Research (95), Market Research and Analysis (88), Probability Theory and Mathematical Statistics (87)

Awards: Second Prize, Fujian Province Challenge Cup Innovation Competition; Project selected for *Leaders' Reference* by Fujian Provincial Government

Activities: Student Union Media Center; Volunteer for the 2023 China International Fair for Investment & Trade (CIFIT)

Research Experience

Large-Scale Network Analysis and GIS-Based Simulation of Lifestyle Factors and Depressive Symptoms Jan 2026 – Present

- Analyzed 105,138 survey records using a 36-node Ledoit–Wolf shrinkage partial-correlation network to characterize conditional associations among lifestyle behaviors, work conditions, demographic factors, and depressive symptoms.
- Conducted sex- and age-stratified network comparisons across 595 pairwise edges using bootstrap-based simultaneous uncertainty bands; implemented a 60% discovery and 40% holdout design together with data-quality and threshold sensitivity analyses to evaluate robustness.
- Identified 13 direct depressive-symptom associations at $|r| \geq 0.04$; all retained the same direction in the holdout sample and 12 remained above the prespecified display threshold.
- Extended the empirical behavioral-network structure into a GIS-based Agent-Based Modeling framework to explore spatially targeted intervention scenarios and effectiveness–equity trade-offs.

LLM-Assisted Agent-Based Modeling Review and Literature Intelligence Workflow Jul 2026 – Present

- Constructed and analyzed a corpus of 29,139 Agent-Based Modeling publications from Web of Science, PubMed, and DBLP, with metadata and abstract enrichment from OpenAlex and systematic deduplication and screening.
- Designed an LLM-assisted literature classification pipeline to identify research domains, modeled agent types, methodological relevance, LLM adoption, and functional roles from paper titles and abstracts, enabling large-scale topic and methodology assessment.
- Conducted longitudinal topic analysis, statistical trend testing, co-authorship network analysis, and Louvain community detection to examine the evolution of ABM research themes, collaboration structures, and the emergence of LLM-enabled agent research.

- (In progress) Developing a retrieval-augmented generation (RAG) workflow that combines literature meta-data, semantic retrieval, temporal and citation information, and structured evidence synthesis to automatically generate research briefs and structured prompts for downstream research agents.

Other Research Experience – Multi-Omics and Biomedical Data Science

- Developed multi-omics and transcriptomic analysis workflows using GEO, TCGA, and GTEx datasets, with experience in differential expression analysis, WGCNA, functional enrichment, machine learning feature selection, survival analysis, and Mendelian Randomization.
- In a gastric cancer and gastrointestinal stromal tumor study, integrated GEO microarray datasets, identified 581 shared hub genes, and applied Random Forest and LASSO feature selection to derive 14 candidate biomarkers including HSD11B1 and CD44.
- In a cervical cancer transcriptomic study, integrated TCGA-CESC ($N = 309$) and GTEx ($N = 19$) RNA-seq data, retained 14,794 high-confidence genes, evaluated multiple machine learning models, and conducted Kaplan–Meier and multivariable Cox analyses across OS, DSS, and PFI endpoints.

Project Experience

- Campus Stray Cat Recognition App**
- Apr 2026 – May 2026
- Developed a CNN-based image classification pipeline using PyTorch and transfer learning, with WeightedRandomSampler to mitigate class imbalance.
 - Trained and evaluated models for seven common campus stray-cat categories plus an unknown category.
 - Built an interactive Gradio interface for real-time image recognition.

Internship Experience

- Business Analyst Intern, Shanghai LinkCare Information Technology Co., Ltd.**
- Aug 2024 – Nov 2024
- Analyzed 50,000+ medical product sales records across 21 provinces; supported performance monitoring, market trend analysis, and supply-chain optimization using Python and BI tools.
 - Conducted procurement and market research involving government institutions and hospitals to support market analysis and business decisions.

Technical Skills

- **Programming:** Python, R, SQL
- **Data Science:** Machine Learning, Network Analysis, Agent-Based Modeling, Natural Language Processing, RAG, Survival Analysis, Multi-Omics Analysis
- **Spatial Analysis:** GIS, Spatial Data Analysis, Spatial Visualization
- **Software:** Microsoft Office Suite, SPSS, Stata
- **Language:** IELTS 7.0 (Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.0)

Awards & Honors

- Second Prize, Fujian Province Challenge Cup Innovation Competition
- Project selected for *Leaders’ Reference* by Fujian Provincial Government

Personal Summary

Interdisciplinary researcher with training in sociology, public health, data science, and computational modeling. Experienced in large-scale observational data analysis, network science, agent-based modeling, LLM-assisted literature analysis, multi-omics data integration, machine learning, and spatial analysis. Interested in combining empirical data, computational social science, and intelligent research workflows to study complex health and social systems.